

Fruit DNA: Extract, Observe, Evaluate

LAUNCH • Science • 40 minutes

I can explain the steps in fruit-DNA extraction, record an observation and evaluate the evidence cautiously.

Prepare before pupils arrive

Setup: Pre-chill ethanol, prepare all solutions and decide whether the class uses physical, adult-demonstrated or parity-data route. Label this NON-CORE.

Safety: Follow local COSHH/risk assessment. Adult controls ethanol throughout; no ignition source, tasting or direct pupil contact.

Equipment: Adult-prepared fruit, salt-detergent solution, sealable bag, filter/mesh, clear tube, cold ethanol controlled by an adult, tray and observation sheet.

Safety: Local risk assessment required. Ethanol is flammable and adult-handled only; keep away from flames/heat. Wear eye protection as locally required. Do not taste or touch materials; wash hands and clear spills promptly.

Fallback: The pupil directs each method step while an adult handles all liquids, or analyses the parity dataset. Direction, observation and evaluation remain the assessed Science.

Access and participation

Offer reading aloud, pointing, signing, quiet thinking, a no-touch route and exact scribing where these support the learner. Keep the same subject decision. In shared work, let each learner commit before feedback; use a partner as card mover or checker, then swap. Avoid public speed rankings.

Source lesson curriculum link: Explain how DNA can be extracted from fruit; connect the observation to genetic information.

40-minute teaching sequence

Stage / total time	Teaching move	Adult support / response
Arrival • 3 min	Describe complementary base pairing. State the chromosome-DNA-gene relationship.	Wait, regulate and retrieve through the lightest workable route. If recall is inaccessible, point to one retrieval sentence; do not teach today's answer.
Starter • 3 min	Think first. Point, say, sign or write your choice.	Protect curiosity: receive every first idea without judgement. Ask 'What makes you think that?' and preserve the prediction for later evidence.

Stage / total time	Teaching move	Adult support / response
I do · 4 min	filtrate — liquid through the filter precipitate — material appearing from solution extract — remove from a mixture Mash the fruit with salt-detergent solution: detergent disrupts cell and nuclear membranes, while salt helps DNA strands clump together. Filter the mixture; collect the filtrate while larger fruit solids remain behind.	Let the teacher/model carry the explanation; pupils watch, point or track. Use a visual cue only if access is the barrier, then fade it.
We do · 4 min	Commit to an answer. Show the evidence you used.	Scan every response mode. Do not infer understanding from handwriting or confidence. Secure: transfer. Mixed: contrast. Misconception: counterexample then a fresh question.
I do 2 · 3 min	Observation: describe colour, location and texture without writing 'pure DNA'. Inference: the extraction method is expected to produce DNA-containing material, but appearance alone cannot identify every component.	Model one complete evidence-to-explanation sentence, then pause. Keep the science words; change density, modality or adult role before changing the goal.
We do 2 · 4 min	Classify each statement after the practical or parity dataset, then repair any overclaim. METHOD OBSERVE INFER LIMIT CONNECT Use the numbered cards in your pupil pack.	Protect pupil operation and thinking; support handling only where needed. Ask what changed and what stayed the same before supplying a scientific word.
Independent · 15 min	Carry out or direct the non-core fruit-DNA extraction, record the result and evaluate the evidence. Record: An ordered method, precise observation, DNA/chromosome/gene connection and one valid evaluation point.	Protect independence. Accept equivalent evidence routes and scribe exact meaning. Prompt only as much as needed; fade as soon as the pupil continues.
Exit · 4 min	Why is 'DNA-containing precipitate' better than 'pure DNA'? Point to, read or explain the evidence you made.	Genuinely receive one final response and name back the Science heard or seen. Give one next move; the pupil edits or re-attempts on the existing work.

The timing belongs to the whole stage. Several PowerPoint slides may share it; do not restart that time on every slide. Full source-stage text is in the PowerPoint speaker notes and original HTML.

Answers, evidence and feedback

Hinge answer: To disrupt cell and nuclear membranes so DNA can be released.

Yes. Detergent disrupts lipid membranes, helping release cell contents including DNA.

Misconception repair

Repaired. The result is DNA-containing precipitate, not purified DNA or a visible individual gene.

Guided checkpoint

Yes. Detergent disrupts lipid membranes, helping release cell contents including DNA. Think about the barriers around the cell and nucleus.

White strands

DIRECT OBSERVATION. Observation: this records what was seen without identifying purity. Ask whether the senses recorded this directly.

DNA-containing material

SCIENTIFIC INFERENCE. Inference: this connects method knowledge to the observation. Ask whether this conclusion uses scientific knowledge as well as sight.

Purity unknown

METHOD LIMITATION. Limitation: appearance alone does not establish purity. Ask what the method cannot establish from appearance.

Model limitation

The precipitate is not purified DNA, yield is not measured accurately by appearance and the activity cannot show a gene, base pair or double helix directly.

Independent evidence

An ordered method, precise observation, DNA/chromosome/gene connection and one valid evaluation point.

Supported: Order four method pictures, select the observed result and complete one limitation sentence. Use release, filter, precipitate and not pure. An adult handles all materials.

Standard: Record method and observation, explain two reagent/separation roles and evaluate one limitation. Use Observation → Inference → Limitation headings.

Stretch: Compare practical/parity outcomes, identify variables that affect visible yield and propose one fair improvement. Do not claim molecular identity or purity; no Higher-tier analytical chemistry is needed.

Record the teaching response

What the learner actually showed	Next teaching action
Secure explanation with evidence	Reduce content prompting; offer another example or a limitation.

What the learner actually showed	Next teaching action
Partly correct / mixed	Point to the deciding evidence and invite a revision.
Access barrier	Change the reading, sensory or response format; keep the subject decision.
Missing source or observation	Keep the gap visible. Name the source or authorised check needed.

Safety and evidence boundary

Use the centre's authorised practical or fieldwork method and local risk assessment. Supplied sample data, fictional place models, card feedback and rehearsals are teaching material. They are not the learner's practical observations or proof of a qualification outcome. For portfolio use, check the exact registered unit/version, accepted evidence and centre process.

Sources and companion notes

Primary classroom source: SCI_L_W12L3_Fruit_DNA_Evidence_Do.html

The uploaded lesson is included unchanged. Text and teaching steps in the PowerPoint are editable; source diagrams are placed images. Word booklets are editable. Static PDFs cannot reproduce HTML controls; use the paper cards/tables or open the original HTML.

Verification scope: matched to the uploaded title, pathway, objective, stage sequence and 40-minute timing. Embedded workbook references are retained as source locators; this is not a new line-by-line audit of every scheme-of-work row or a centre assessment sign-off.